Standing Waves

> SYNOPSIS
>
> We look at the conditions for forming a standing wave and derive equations describing the standing wave.

> BEFORE YOU READ ON — PREDICT
>
> A guitar string, clamped at both ends, plays one definite note. Three things you could change: **fret it, so the vibrating part is shorter**, **tighten the tuning peg**, **pluck it harder**. Which of those change the *note*? Write down your three answers. By the end of this note all three fall out of one formula.

PERFORMANCE EXPECTATIONS

1. **Add** two equal waves travelling in opposite directions to obtain $2A_0 \sin kx \cos\omega t$.
2. **Explain** why that product is a standing wave: which factor carries position and which carries time.
3. **Locate** the nodes and antinodes from $\sin kx = 0$ and $\sin kx = \pm 1$.
4. **Apply** the fixed-end boundary conditions to reach $\lambda_n = 2L/n$.
5. **Compute** the frequency of the nth harmonic from the string's tension, length and mass per unit length.

1. Standing Waves

We start with a moving wave (forward going)

$$A_+(x, t) = A_0 \sin(kx - \omega t)$$

If we pluck a guitar string (fixed at both ends) we launch a moving wave. If we continue to strum, we launch multiple waves, all forward moving.

★ **What happens when the forward going wave reaches the fixed end?**

They bounce back — 180° out of phase with respect to the incoming wave — and start to move in the opposite direction, so the returning waves are

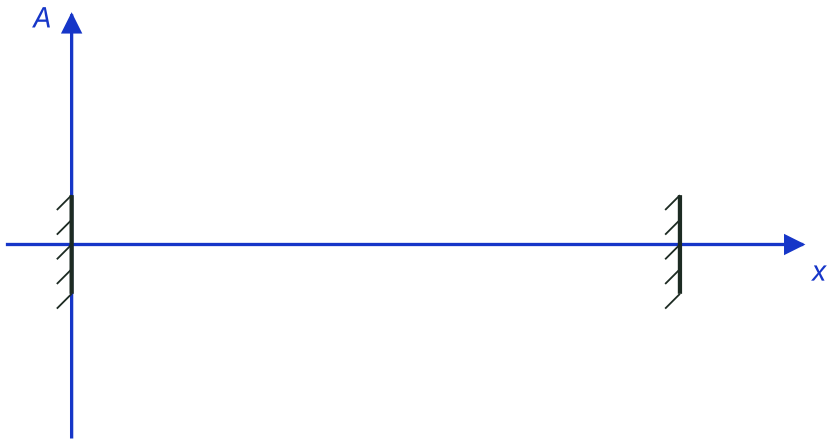
$$A_-(x, t) = A_0 \sin(kx + \omega t)$$

The overall effect of the interference between the forward and the returning waves is that they create the **standing wave**.

1. Write the forward-going wave, and then the returning wave after it has reflected off the fixed end.
2. The string carries both at once. Write the total displacement as a sum.

3. Which trig identity turns that sum into a product? Use it, and see what cancels.

4. Look at what is left. Which factor depends only on x, and which only on t? What does that say about where the zeros are?



Draw the standing wave at full amplitude here, then two or three later moments. Mark every point that never moves.

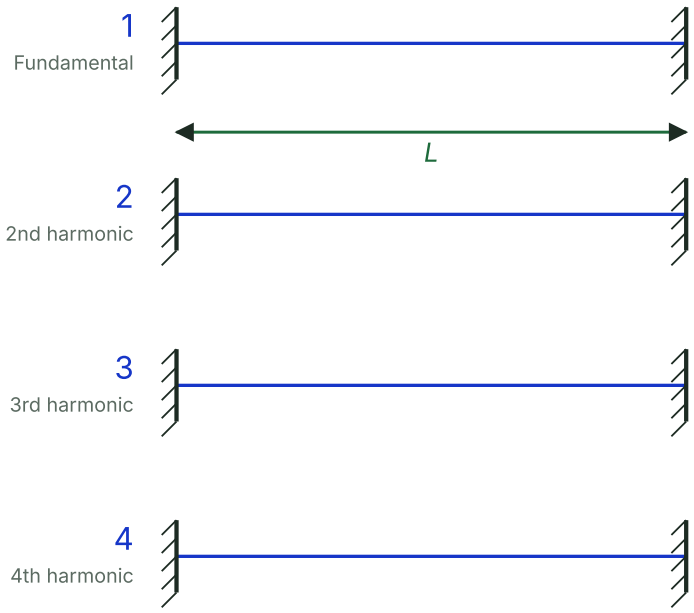
Standing waves on a string with 2 fixed ends

The speed is fixed by the string itself, not by us:

$$v = \sqrt{(T/\mu)} \quad \text{and} \quad v = \lambda f$$

so once the string is strung and tuned, choosing a wavelength *chooses* the frequency. The clamped ends decide which wavelengths are allowed.

5. The string is clamped at $x = 0$ and at $x = L$. What must the standing wave do there, for every t? What does that force on $\sin kx$?
6. Sketch the first four shapes that satisfy it, and read the wavelength of each off the picture in terms of L.
7. Use $v = \lambda f$ to turn each wavelength into a frequency. Then write λ_n and f_n for the nth.
8. v is a property of the string. Of its length, its tension, and how hard you pluck it, which ones change f_n?



Draw the first four allowed shapes between the clamps. Mark the nodes, and write the wavelength of each beside it in terms of L.

THE PUNCHLINE

A string clamped at both ends cannot vibrate at just any frequency. It has a **discrete** set, evenly spaced, and every one is a whole-number multiple of the lowest. Nothing in this argument is about guitars: it is what happens to *any* wave shut into a box, and the same counting comes back for a pipe of air, a microwave cavity and an electron in an atom.

Looking ahead

You now have the shapes a trapped wave is allowed to take. **Note 10** asks what it is carrying: the energy in an oscillation, where it sits at each moment, and the speed of the oscillator at any point of its swing.

Where this sits. Note 7 built one travelling wave, $f(x - vt)$, and got its speed out of the string. This note sends two of those at each other. The sum of two travelling waves is not a travelling wave at all — x and t come apart, the shape stops moving, and the clamps then throw away all but a discrete list of it.

BACK TO YOUR PREDICTION

Fret it: higher. L is underneath $f_n = nv/2L$, so halving the vibrating length doubles every frequency — an octave, which is why the 12th fret sits at the middle of the string. **Tighten it: higher.** T is on top of $v = \sqrt{(T/\mu)}$ and v is on top of f_n. **Pluck harder: no change.** Louder, same note — the amplitude appears nowhere in λ_n or f_n. That is the same shape of answer as Note 7 gave for the speed of a pulse, and for the same reason: the medium sets the terms.